SilicaGuard

AI-Powered Silicosis Screening & Prevention Platform

For Zimbabwe's Artisanal Gold Mining Communities: Kwekwe, Midlands Province

Cimas Healthathon 3.0 · Innovation Submission

Team Name	SilicaGuard
Team Lead	Panashe M Chandiwana: AI Engineer
Solution Type	HealthTech: AI Screening + Clinical Decision Support + Population Health Analytics
Target Location	Kwekwe District, Midlands Province, Zimbabwe
Channels	Android tablet (offline) · USSD · WhatsApp · Web dashboard
Tech Stack	Python FastAPI · React Native · Claude AI · DenseNet-121 CNN · PostgreSQL ·

The crisis at a glance

~1/week Silicosis death at Kwekwe District Hospital (Dr T. Gunda, 2025)	**50–60** Confirmed annual silicosis deaths in a single district	**19%** Silicosis prevalence among Zim artisanal miners (peer-reviewed)	**0** Existing digital screening tools for silicosis in Zimbabwe

1. Problem Statement

Silicosis is a fatal, irreversible occupational lung disease caused by prolonged inhalation of fine crystalline silica dust generated during gold mining, rock drilling, blasting and ore crushing. Once silica particles enter the lungs they trigger permanent fibrosis, progressive scarring that destroys breathing capacity over months or years with no cure available. The disease is entirely preventable, yet it continues to kill at a rate that constitutes a public health emergency in Zimbabwe's mining communities.

The crisis is concentrated most acutely in Kwekwe, Midlands Province, Zimbabwe's most densely mined gold town. Kwekwe District Hospital maintains between 12 and 15 silicosis inpatients at any given time and its Medical Superintendent, Dr Tinashe Gunda, confirmed publicly that the facility records approximately one silicosis death per week thus leading 50 to 60 deaths per year. He described this figure as "UNACCEPTABLE" and explicitly called for community-based digital screening tools and health education outreach as the only viable path to reducing the burden. No such tool currently exists in Zimbabwe.

> "We average about one death a week. That gives us 50 to 60 deaths a year, which is unacceptable. What we need is health education and outreach. We must go into mining communities to screen workers." Dr Tinashe Gunda, Medical Superintendent, Kwekwe District Hospital, 2026

The affected population is Zimbabwe's artisanal and small-scale gold mining (ASGM) community estimated at 500,000 to 1.5 million people nationally who delivered 75% of Zimbabwe's record 46.7-tonne gold output in 2025. Peer-reviewed research conducted specifically in Zimbabwe's Midlands Province found silicosis prevalence of 19% among 3,821 artisanal miners, with a mean age of only 35.5 years. The same population carries a tuberculosis prevalence of 6.8% and HIV prevalence of 18%, creating a deadly triple burden: silicosis raises a miner's TB risk by three to five times and HIV-positive miners are 1.25 times more likely to develop silicosis. These are young men and women losing their lung function irreversibly, in their most economically productive years, with no early warning system to tell them it is happening.

The structural gap at the heart of this crisis is the complete absence of any systematic, community-facing digital screening pathway. Miners have no way to assess their own risk. Village Health Workers have no instrument to identify and refer high-risk individuals. Healthcare professionals at primary care level have no decision support tool for occupational lung disease. By the time a miner presents at Kwekwe District Hospital, the disease has typically reached Stage 2 or Stage 3 at which point management is purely palliative. The entire window for preventive intervention, the years between first significant exposure and clinical crisis is unmonitored, unmanaged and invisible to the health system. SilicaGuard is built to make that window visible and to act within it.

2. Proposed Solution

SilicaGuard is a HealthTech platform that delivers AI-powered silicosis risk screening, intelligent referral, health education and occupational exposure surveillance directly to Zimbabwe's artisanal mining communities through the channels miners already have. It operates across three interfaces sharing a single AI backend: a tablet-based field screening tool used by healthcare workers during mine site outreach, a USSD channel accessible from any mobile phone with/without internet and a web-based analytics dashboard for healthcare facilities and insurers.

The healthcare worker's role(the human at the centre)

Primary Clinical Role: Nurse / Clinical Officer / Village Health Worker (VHW)

- Carries the Android tablet to the mine site and administers the 10/more-question screening interview reading each question aloud so the miner never needs to read or touch the device.
- Reviews the AI risk classification on-screen and explains the result to the miner in plain language, answering immediate questions.
- For Refer Now cases: hands over the QR-coded referral card and explains what will happen at the hospital. The system has already pre-alerted Kwekwe District Hospital.
- Monitors the dashboard for USSD self-screens in their catchment area and initiates follow-up with high-risk miners who have not yet attended a facility.
- Plays a critical role in community trust SilicaGuard provides the intelligence; the healthcare worker provides the human relationship that makes miners act on it.

The clinical entry point is the SilicaGuard field screening tool. The VHW reads 10/more structured questions covering years worked underground, job role, dust suppression practices, PPE use, cough duration, breathlessness graded on the Medical Research Council dyspnoea scale, TB history, weight loss, chest pain and prior lung diagnosis. On completion, the AI risk engine powered by the Claude API with a Zimbabwe-specific clinical system prompt processes all these responses and produces a colour-coded classification (Low Risk / Watch / Refer Now) with a plain-language explanation in Shona or English. For Refer Now cases, the app generates a QR-coded digital referral card and simultaneously pre-alerts Kwekwe District Hospital. The entire screening operates fully offline with automatic sync when connectivity returns.

The USSD self-screening channel extends reach to miners who cannot attend an outreach day. Any miner dials a shortcode from any phone on any Zimbabwean network no data, no smartphone required. Six/More questions complete in under three minutes. Within 60 seconds of ending the call, a WhatsApp message delivers the AI-generated risk result, explanation in Shona and nearest clinic details. Following any screening contact, every miner is enrolled in a four-week WhatsApp education campaign in Shona, Ndebele and or English, covering silicosis facts, correct N95 use, early warning signs and the miner's legal rights under the NSSA occupational disease compensation programme.

When a referred miner attends Kwekwe District Hospital for a chest X-ray, SilicaGuard's AI imaging module enters the clinical workflow. The X-ray image is uploaded to the platform. A deep learning computer vision model such as (DenseNet-121 architecture pre-trained on 100,000+ chest radiographs) and fine-tuned on a 49,872-image annotated pneumoconiosis dataset classifies the image as Normal (Stage 1 silicosis), or Stage 2 or 3 silicosis within 60 seconds. It also generates a Grad-CAM heatmap overlay on the X-ray, showing the clinician exactly which lung regions drove the AI's classification. The clinician makes the final determination. SilicaGuard gives every overworked district hospital doctor an immediately available, research-validated second opinion on every film without waiting for a specialist.

Technology stack (Proposed)

Tablet screening app	React Native · Offline SQLite · Auto-sync · QR referral generator
AI risk engine	Claude API (claude-sonnet-4-6) · Zimbabwe clinical system prompt · Shona/Ndebele output
X-ray AI model	PyTorch · DenseNet-121 transfer learning · Grad-CAM heatmap · Runs locally on server
WhatsApp AI agent	Meta WhatsApp Cloud API · Claude API conversational agent · 24/7 Shona-language responses
USSD channel	Africa's Talking · Covers Econet, NetOne, Telecel · Zero data cost to miner
Analytics dashboard	React · Recharts · Leaflet.js district map · AI narrative generator · PDF/CSV export
Backend	Python FastAPI · PostgreSQL · Redis · Hetzner VPS hosting
Data & privacy	Phone numbers hashed · Symptoms anonymised · Zimbabwe Cyber & Data Protection Act 2021

3. Expected Impact

SilicaGuard's measurable outcomes operate at three levels individual miners, the healthcare system and the Cimas member population with quantified targets tracked in real time on the analytics dashboard and visible simultaneously to SilicaGuard, Kwekwe District Hospital, and Cimas.

19/100 Expected silicosis cases identified per 100 miners screened based on published prevalence	**70%** Target referral completion rate: Refer Now cases confirmed as completing X-ray within 6 weeks	**20–30%** Target reduction in Stage 2–3 admissions at Kwekwe District Hospital within 18 months	**30%** Target increase in consistent N95 use among enrolled miners at 6-month rescreening

At the individual level, applying the published 19% silicosis prevalence to a 100-miner outreach day yields approximately 19 miners who would otherwise remain undiagnosed until symptomatic crisis. The referral pre-alert mechanism targets 70% referral completion meaning 70% of Refer Now miners will be confirmed as having completed a chest X-ray within six weeks of screening, compared to the near-zero follow-up rate in the current unstructured pathway where miners receive verbal advice and return to the shaft. For every miner identified at Stage 1 and removed from high-exposure work, progression to the hospitalisation-level disease that fills Kwekwe's ward is halted or slowed. These are preventable deaths.

At the health system level, systematic community screening is expected to reduce new Stage 2 and Stage 3 silicosis admissions to Kwekwe District Hospital by 20 to 30% within 18 months of full deployment a projection grounded in outcomes from South Africa's formal mining sector when periodic silicosis screening was introduced. For Cimas, the measurable outcome is a direct reduction in high-cost respiratory hospitalisation claims from the mining sector, quantifiable against existing claims data. The AI-generated exposure profile for each insured mining company gives Cimas's actuarial team risk intelligence, enabling proactive wellness intervention before claims materialise. This is SilicaGuard's HealthTech commercial value: turning occupational health data into insurance intelligence.

At the population level, the four-week education campaign targets a 30% increase in consistent N95 use at six-month rescreening a leading indicator that reduces incidence over the coming decade regardless of curative capacity. Beyond Kwekwe, the model scales with no reengineering to Kadoma, Bindura, Battlefields and Shamva. At 19% prevalence across an estimated 60,000 ASGM miners in Zimbabwe's four hotspot provinces, over 11,000 individuals are estimated to carry silicosis today without knowing it. SilicaGuard's long-term ambition is to become the national digital infrastructure for occupational lung health surveillance in Zimbabwe and to replicate across Sub-Saharan Africa's ASGM belt.

4. Final Product

SilicaGuard's final state is a unified three-interface HealthTech platform that functions simultaneously as a community screening tool, a clinical decision support system and a population health observatory, the first of its kind for occupational lung disease in Zimbabwe.

Interface 1: Field screening app (VHW / nurse: Android tablet)

The home screen shows the day's outreach location, miners screened so far and a single prominent "Screen new miner" button. Questions appear one at a time in large text, the healthcare worker reads each aloud and taps the response. Earlier answers automatically increase the clinical weighting of subsequent questions a miner who reports 12 years of dry drilling triggers elevated sensitivity in the symptom questions that follow. On completion, a full-screen colour card displays the risk level in Shona or English with a single action instruction. Refer Now cases generate a printable A6 QR-coded referral card; the system simultaneously pre-alerts Kwekwe District Hospital with the miner's screening record. Fully offline, syncs automatically on reconnection. Fifteen miners screened per morning shift without WiFi.

Interface 2: USSD self-screen + WhatsApp AI agent (any phone, any network)

Dialling the shortcode launches a six-question numbered-menu flow completed in under three minutes at zero data cost. Thirty seconds after the session ends, a WhatsApp message delivers the AI-generated risk classification, plain-language explanation, and nearest clinic details in Shona. The WhatsApp channel is powered by a conversational AI agent available 24 hours a day, miners can ask follow-up questions in Shona/Ndebele/English at any time and receive contextual, clinically grounded responses. The four-week education sequence runs in the background: week one covers what silicosis is, week two correct N95 use, week three early warning signs, week four NSSA compensation rights. The healthcare worker monitors the dashboard for USSD high-risk self-screens and initiates outreach contact.

Interface 3: Hospital / Cimas analytics dashboard (web browser)

The dashboard opens on three real-time headline metrics: total miners screened, percentage high-risk and referral completion rate. Below: a Leaflet.js map of Kwekwe district with mine sites as colour-coded pins (green / amber / red by risk concentration), a referral tracking panel showing every Refer Now miner from the current month with completion status (green = X-ray confirmed, amber = pending, red = not followed up after two weeks), a 12-month silicosis admission trend graph for Kwekwe District Hospital, the chest X-ray AI module for uploading and classifying X-ray images with Grad-CAM heatmap output and a Monday morning AI narrative a paragraph written by the AI summarising the previous week's data and directing the outreach team's priorities for the coming week. All data exports to PDF and CSV for NSSA, MoHCC and Cimas reporting.

> Design principle across all three interfaces: offline-first for reliability single-action-per-screen for usability traffic-light visual language for immediate comprehension. SilicaGuard is built for the shaft entrance at Globe and Phoenix mine at 6am, for the hospital clerk with 30 patients waiting and for the miner who checks WhatsApp at midnight to understand why they cannot stop coughing.